

WSPRadar.org reconstruction & comparison

SNR differences between a short portable vertical and a reference vertical

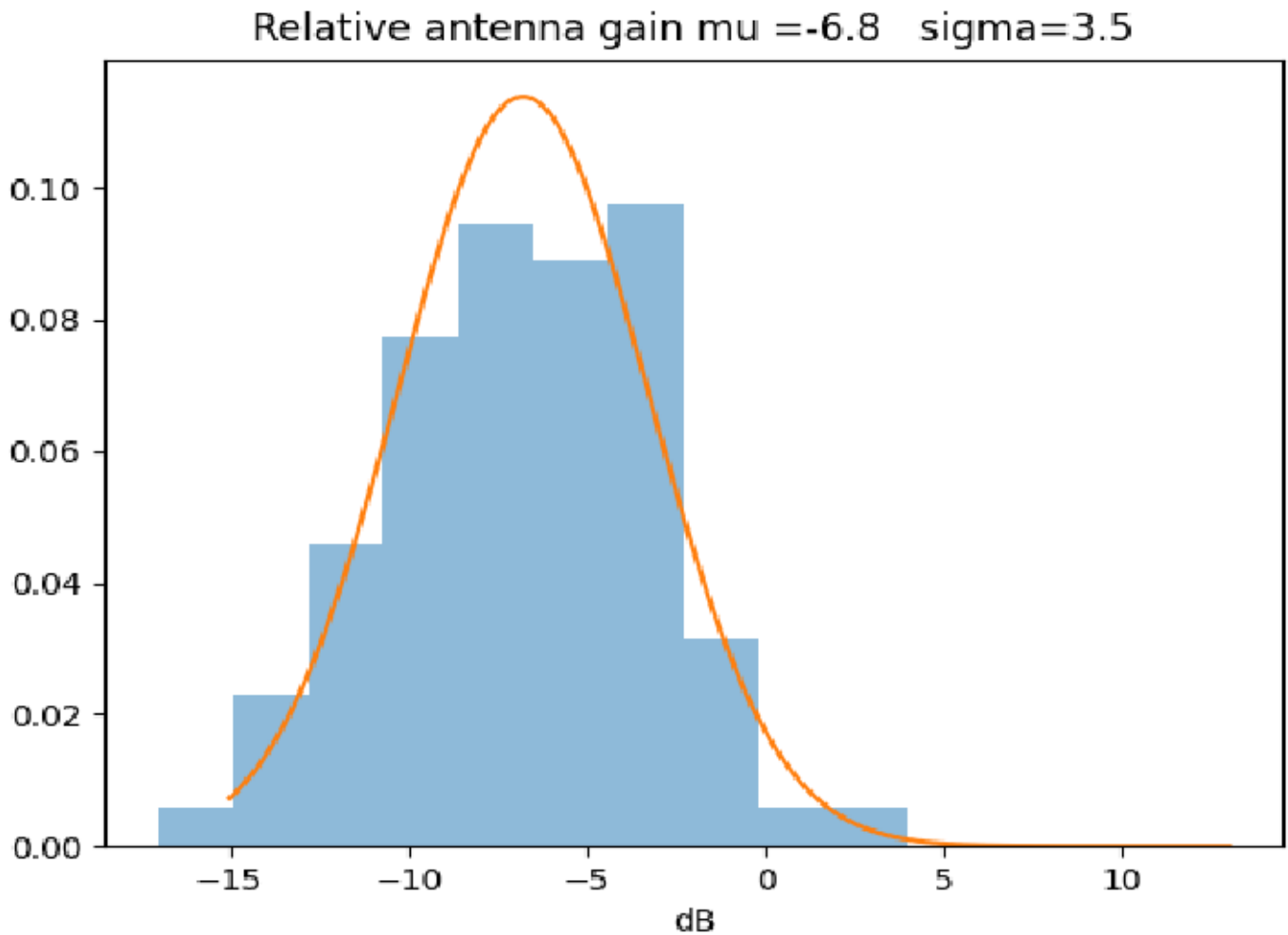
Referenced publication: Zander, J. (2022). *Simple HF antenna efficiency comparisons using the WSPR system*.

Source figure: **Figure 4, p. 4** · arXiv:2209.08989v1 · 19 September 2022

Demo: Experiment A · 21 May 2022, 09:30-10:30 UTC · 20 m (14 MHz) · Target SK0WE/P – Reference SK0WE/1 · JO97

Panel A - Original publication

Original image; orange line: paper's normal curve.



Rounded paper statistics

Mean -6.8 dB · Reported σ 3.5 dB

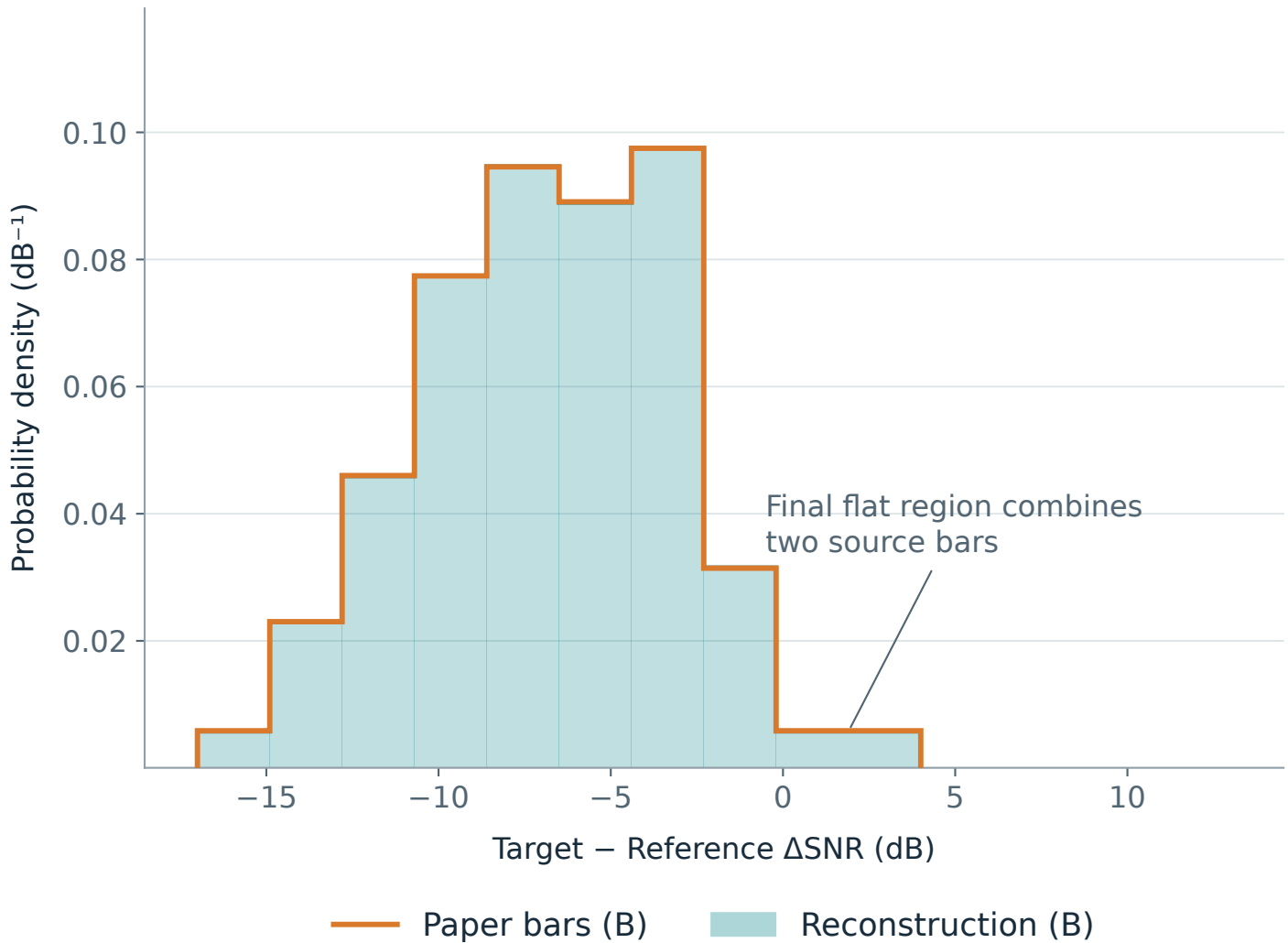
Current WSPRadar reconstruction

Mean -6.8 dB · Sample σ 3.5 dB

166 receiver-cycle pairs · 37 receiver identities · 12 cycles

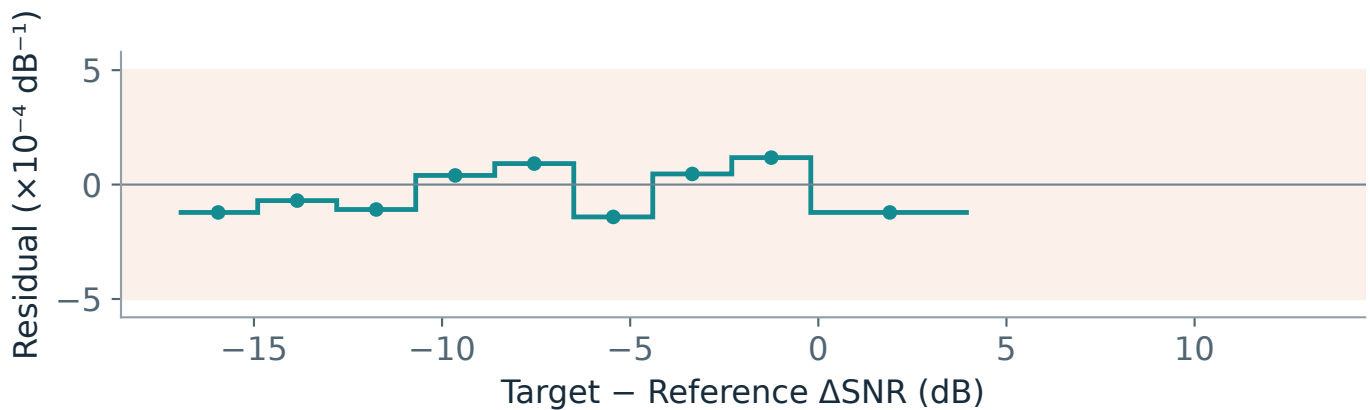
Panel B - Reconstruction

Paper-style bins, with digitized paper bars overlaid.



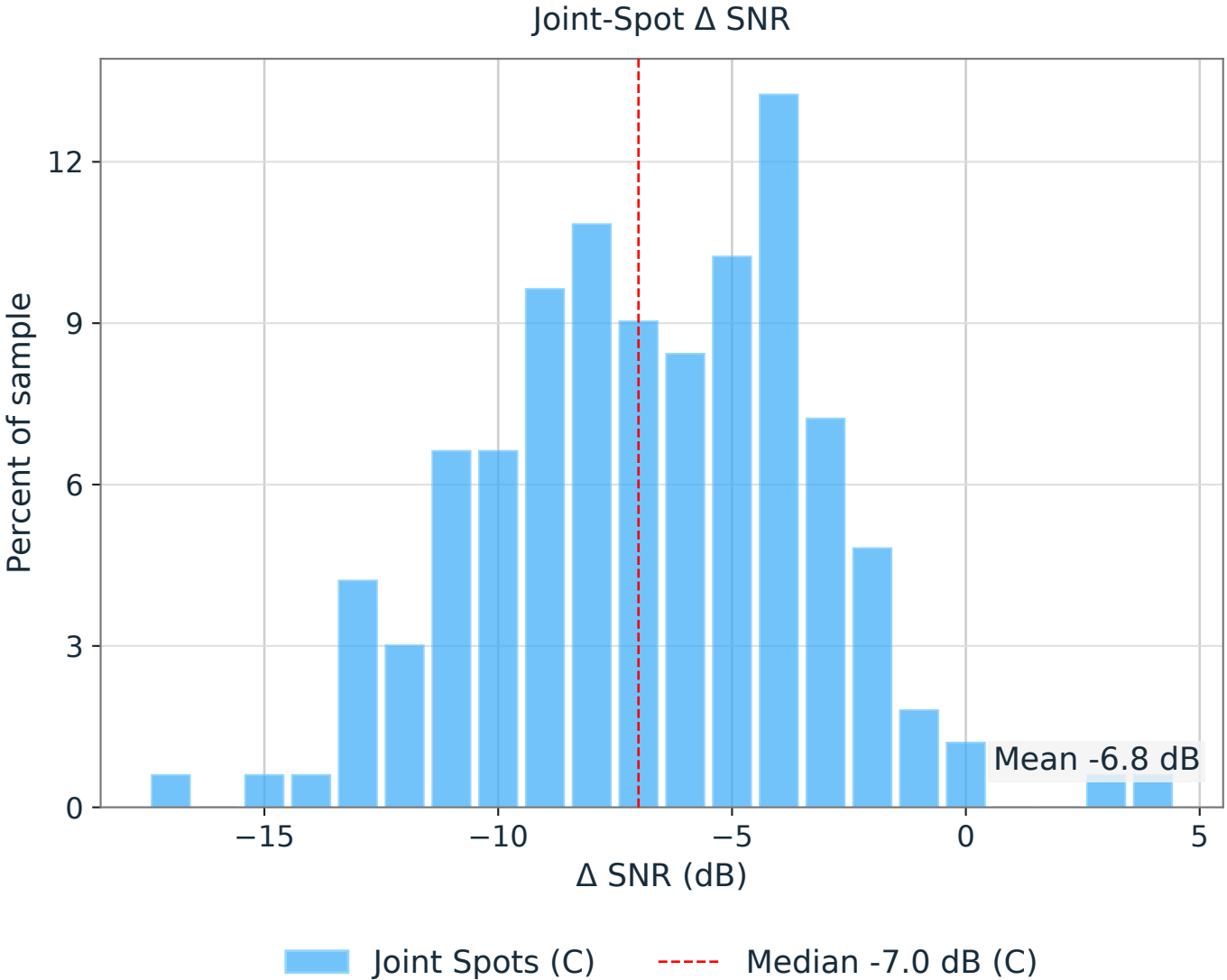
Agreement check - Panel B minus paper

Maximum |residual| 1.41×10^{-4} dB⁻¹



Panel C - WSPRadar view

Native Joint-Spot histogram with 1 dB bins.



Same observations, finer display

The 1 dB bins reveal uneven counts and empty bins that the paper's coarser histogram combines. The mean and standard deviation do not change.

Panel B: density = count / (166 $\times$ bin width in dB); bar areas sum to 1. Panel C: percent of all 166 pairs per 1 dB bin; bar heights sum to 100%.

Read both histogram axes linearly. Panel C retains WSPRadar's native binning, colours and statistics.

No fitted curve, time shift or density rescaling is applied to the reconstruction.

READ Δ SNR = Target – Reference: negative values favour the Reference vertical. Each Joint Spot pairs both signals at one receiver in one cycle.